MATHEMATICS
Introduction to Modern Algebra II
Quiz 7.

Let E be the splitting field of $x^q - x \in \mathbb{Z}_p[x]$ where $q = p^m$, p is a prime and m is a positive integer. Show that $|E| = q$.

Proof. Let R be the set of all roots of $x^q - x$ in E. We claim that $|R| = q$. It suffices to show that $x^q - x$ has no repeated roots. This follows since $\delta(x^q - x) = -1$ and so $(x^q - x, \delta(x^q - x)) = 1$ (recall that if a is a repeated root of $f(x)$, then $(x-a)^2$ divides $f(x)$ and $(x-a)$ divides $\delta f(x)$ and so $(f(x), \delta f(x)) \neq 1$ as it is divisible by $x - a$).

Next we claim that R is a field. Clearly $0, 1 \in R$. If $a, b \in R$, then we have $(a+b)^q = a^q + b^q = a + b$ and so $a + b \in R$; $(ab)^q = a^q b^q = ab$ and so $ab \in R$; $(-a)^q = (-1)^q a^q = -1a = -a$ and so $-a \in R$; if $a \neq 0$, then $(a^{-1})^q = (a^q)^{-1} = a^{-1}$ and so $a^{-1} \in R$. Thus R is a field.

It follows that R is the splitting field of $x^q - x$ and $|R| = q$ as required. $\square$